

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mathematics

2. Subject Name: Numerical Analysis **L-T-P:** 3-1-0 **Credits:** 4

3. Pre-requisites: NA

4. Syllabus and reference books:

Floating point number system, Floating point arithmetic & rounding error, Big “O” & Little “o” notation, Bisection method, Method of False position, Fixed point iteration scheme, Newton’s method, and Convergence analysis; Gaussian elimination, LU decomposition, Gauss-Jacobi method, Gauss-Seidel method, Conjugate gradient method, and Newton’s method for solving non-linear systems of equations; The Weierstrass approximation, Taylor polynomial approximation, Lagrange interpolation, Newton form of interpolation polynomial, Hermite interpolation, Least square polynomial approximation, piecewise linear interpolation and cubic spline interpolation; Trapezoidal Rule, Simpson’s $1/3^{\text{rd}}$ & $3/8^{\text{th}}$ rule, Gaussian quadrature, Monte Carlo & quasi-Monte Carlo methods; Initial value problems, Picard’s methods of successive approximations, Euler forward & backward methods, Mid-point method, Linear multistep methods (LMM), Adams-Bashforth methods, Backward differentiation formulas (BDF), Runge-Kutta methods, and their consistency, zero-stability, absolute stability & convergences, methods for stiff systems, Boundary value problems, Shooting method for numerical solution of BVP, Finite-difference method.

Space for entering the syllabus

Reference Books:

1. Kendall Atkinson, An Introduction to Numerical Analysis, Second edition, Wiley.
2. John A. Trangenstein, Scientific Computing, volumes 1, 2, & 3, Texts in Computational Science and Engineering, Springer.
3. E. Hairer, C. Lubich, and G. Wanner. Geometric numerical integration: structure-preserving algorithms for ordinary differential equations. Springer, 2006.
4. J. D. Lambert. Numerical methods for ordinary differential systems: the initial value problem. Wiley, 1991.
5. R. LeVeque. Finite difference methods for ordinary and partial differential equations. SIAM, 2007.
6. A. Quarteroni, R. Sacco, and F. Salieri. Numerical mathematics. Springer, 2007.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Numerical Sol of Non-linear equations of one variable	7
2.	Direct and Iterative methods for solving linear system	8
3.	Polynomial approximation theory & Numerical differentiation	8
4.	Numerical Integration	5
5.	Numerical solutions of ODEs	12
Total number of hours		40

Note: Around 12 tutorial classes will be taken